📄 Report: Image Processing Toolkit (Task 3)

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Course: AIML - A
Subject: Computer Vision
Task: 3 – Image Processing Toolkit Submission

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1. Introduction

The objective of this assignment is to design and implement an **Image Processing Toolkit** that allows users to upload, process, and visualize images through a **Streamlit-based GUI**. The toolkit provides an interactive way for applying different computer vision techniques.

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2. Tools & Technologies

- **Programming Language:** Python
- **Libraries Used:**
    - Streamlit (for GUI)
    - OpenCV (for image processing)
    - NumPy (for numerical operations)
    - Pillow (for image handling)

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3. Implementation

- **app.py:** Streamlit-based GUI that supports image upload and displays processed results.
- **Jupyter Notebook:** Contains detailed implementation and explanations of image processing functions.
- **Report:** Explains the workflow and functionality.

Features Implemented:

1. Upload an image (PNG/JPEG).
2. Display original image information (size, format).
3. Apply processing operations such as:

- Grayscale conversion
- Blurring (Gaussian/Median)
- Edge detection (Canny)
- Image resizing
- Brightness & contrast adjustments

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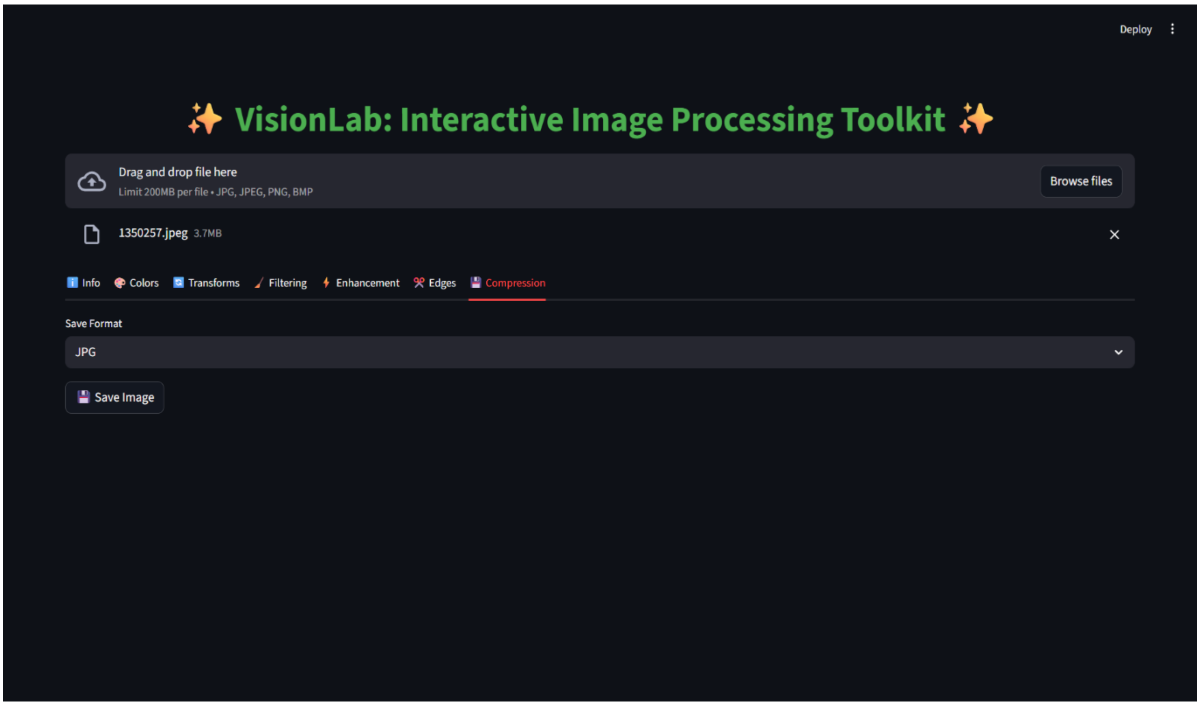
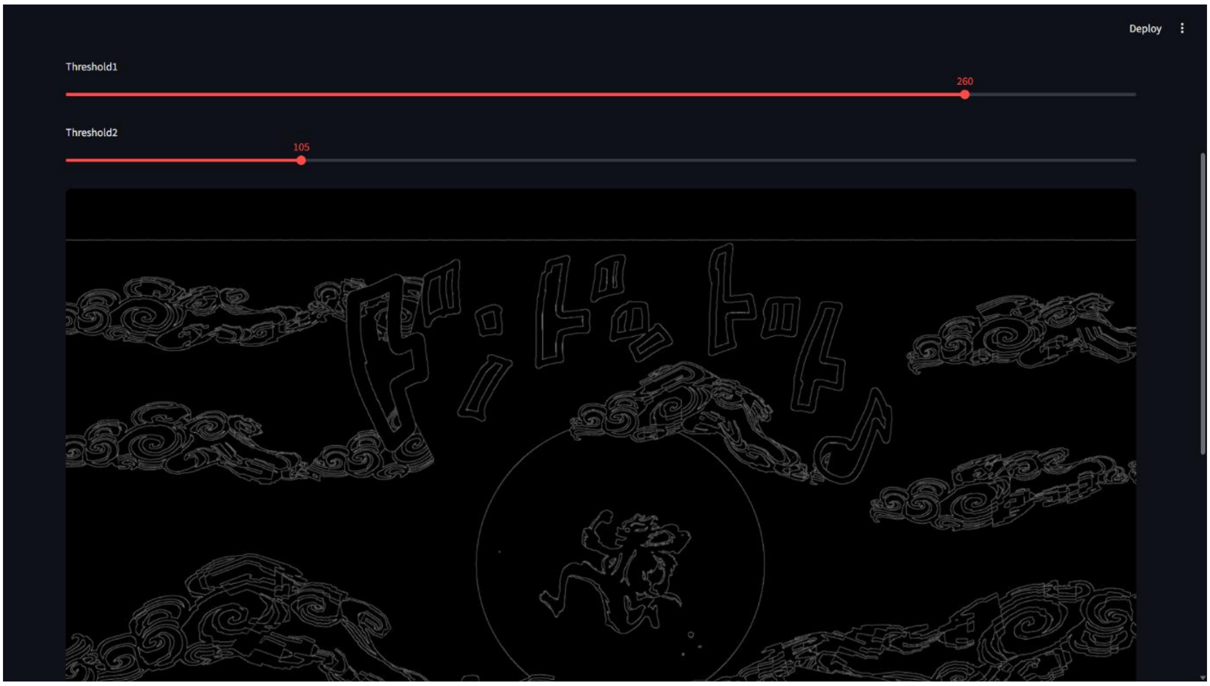
4. Results

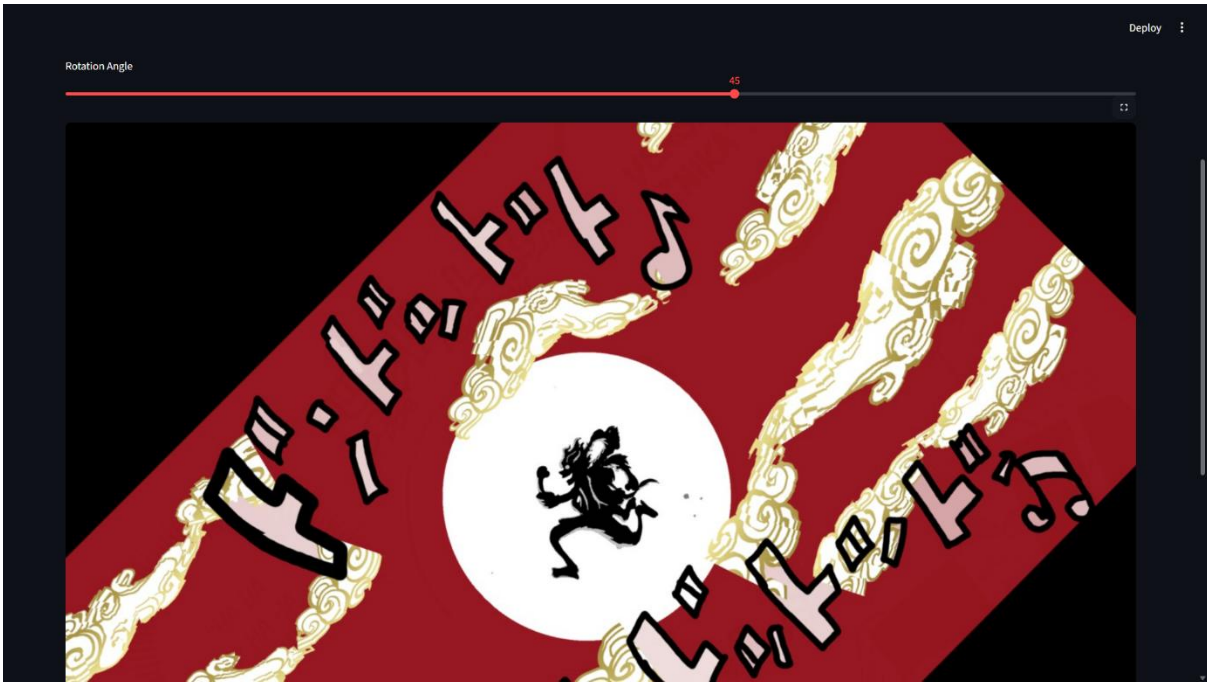
- The toolkit successfully loads and processes different types of images.
- Processed images are displayed side-by-side with the original.
- Users can easily save processed images.

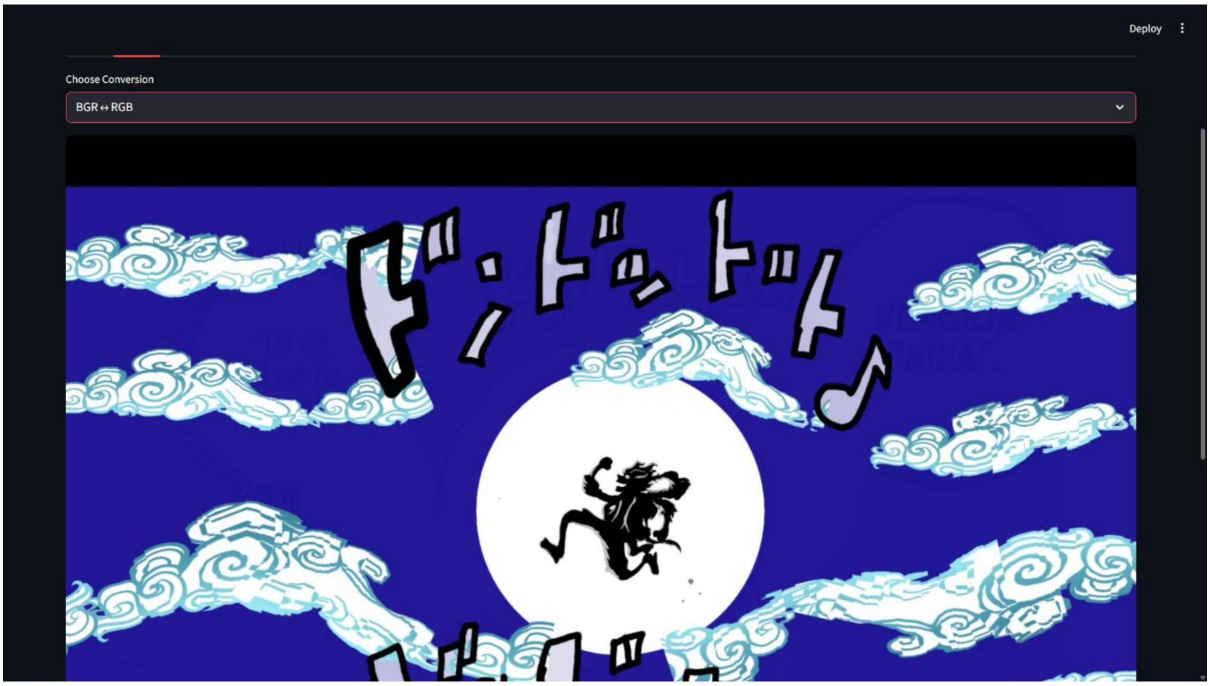


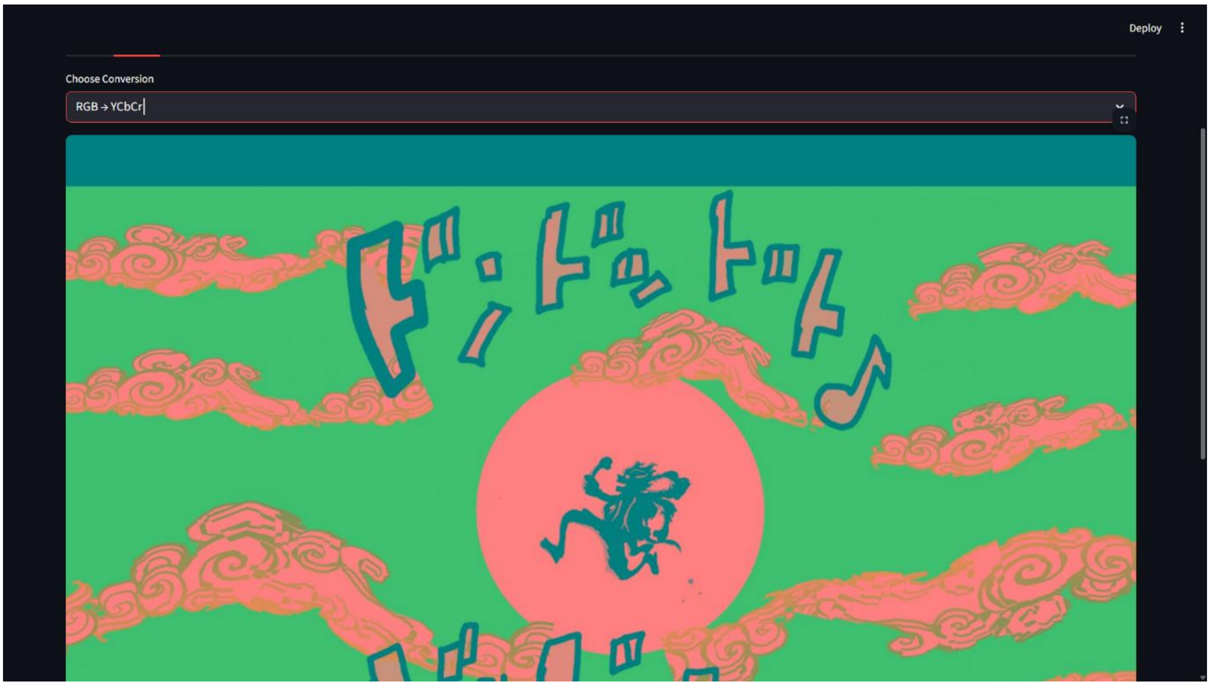

Deploy
Threshold1
260
Threshold2
105

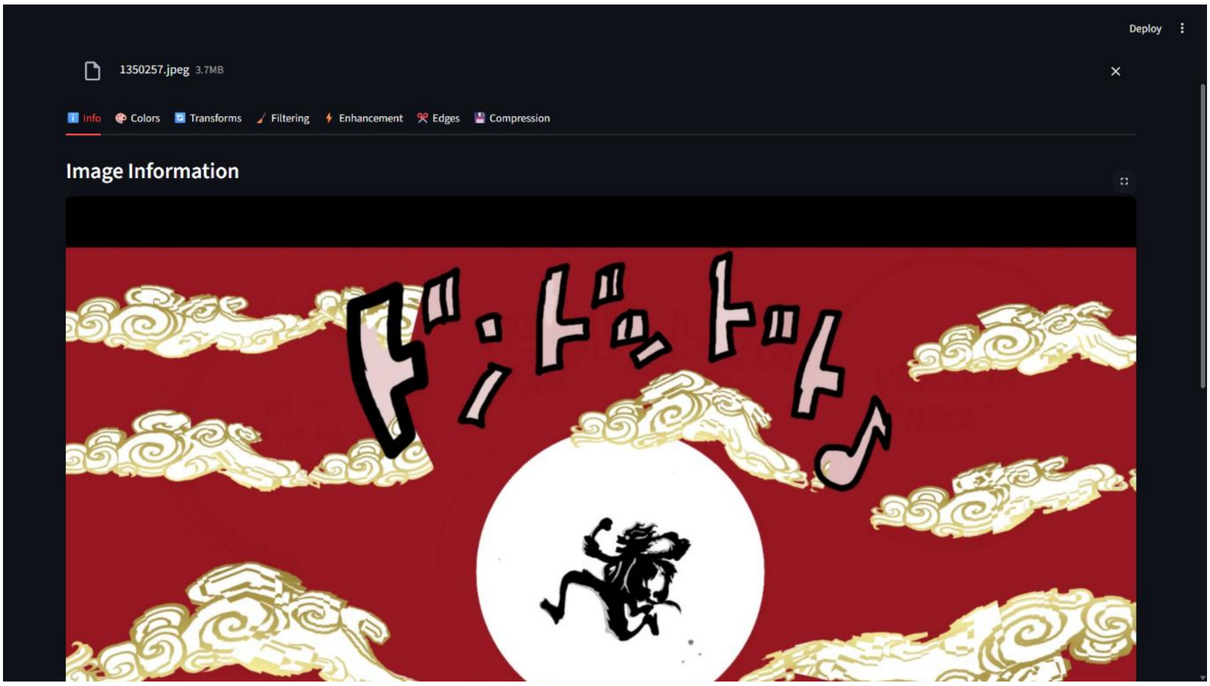

Deploy
Choose Enhancement
Histogram Equalization
DOM

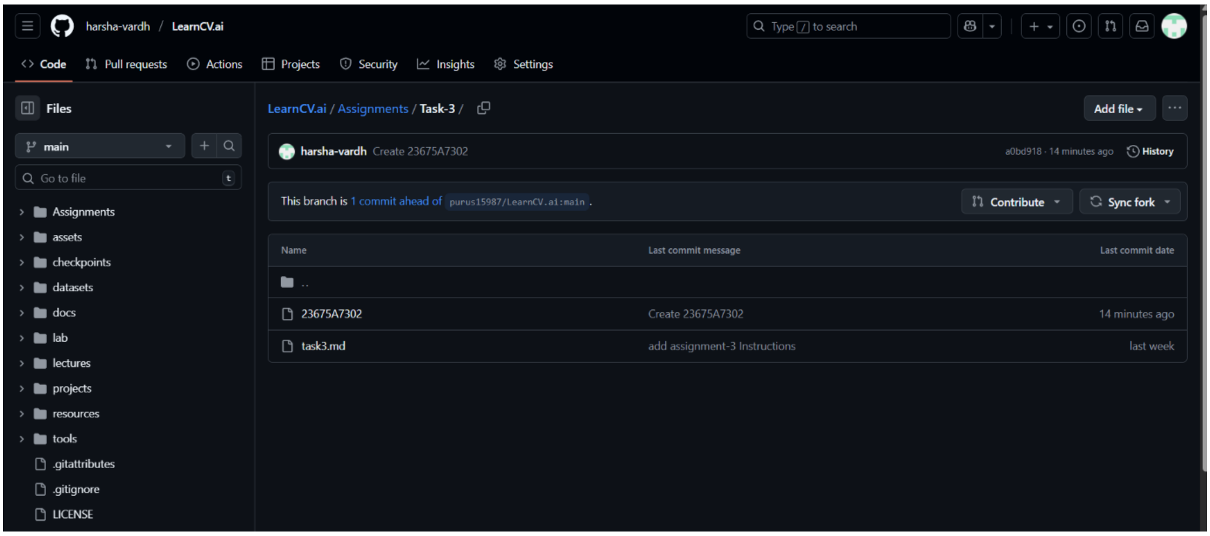

Deploy
Choose Filter
Laplacian


Deploy
Rotation Angle
45


Deploy
Choose Conversion
BGR ↔ RGB


Deploy
Choose Conversion
RGB → YCbCr





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5. Conclusion

The **Image Processing Toolkit** provides a simple and interactive way to experiment with various computer vision techniques. It demonstrates practical usage of OpenCV and Streamlit for building GUI-based applications.